

3 Phases of Object Recognition

Early Attempts - Hand Designed features & models.

Geons
Recognition by Components

↓
Generalized Cylinder

↓
Parts + Springs.

Theoretically beautiful but they don't work.

2nd Phase
Hand designed features, learned models

Pictorial Structures & Constellation Models
Non Parametric Bayes

Boosting
Conditional Random Field
Bay of words
And Or graph.
Datasets - Pascal VOC, Caltech 101
Pop. Biederman - 30K Categories of
(30K - 100K) ~~conjecture~~ Visuals

ImageNet - 15M images & 22k categories

this brought back Convolutional Neural Network - a classic algorithm.

This brought us to 3rd Phase

2012 & Pre-Transformers era.
Screenshots

Vision contributed to deep learning

One could imagine objects would be sufficient to categorize a scene but relationship between objects must be coded.

Scene Graph Representation of visual relationships,

entities-attributes-inter-entity relations

Visual Genome

Compositionality $\rightarrow$ Long tail relations

ex: person riding horse + person wearing
 $\Rightarrow$ horse wearing a hat

ex: person sitting on chair + fire hydrant
 $\Rightarrow$ Person sitting on fire hydrant.
(Zero shot or inference time)